

ARCHIMEDES' APEX

Archimedes' Apex has been carefully structured to establish the importance of Mathematics as the 'language of the universe'. The category tests not just what the participants know, but how well they can apply it. Spanning multiple mathematical concepts, it is ensured that no single area of expertise dominates, and that teams must demonstrate versatility, strategic thinking, and collaboration.

Round 1: The Frequency Analysis

ROUND DURATION: 1.5 HOURS

DELEGATE CAP: 3

Encryption is a complex art, based solely on the fundamentals of Mathematics. However, having a decrypted message is not enough: the ability to understand that message is essential for true success in the field. For this round, participants will be expected to have knowledge of basic ciphers as well as frequency analysis to decrypt messages. Participants should also be aware of Number Theory, Patterns, Geometry, and other basic mathematical domains to reach the final answers.

Round 2: The Triangulation Race

ROUND DURATION: 2-3 HOURS

DELEGATE CAP: 4

GPS tracking, navigation systems, Radars and Sonars and other every day navigational systems find their roots in the depths of Mathematics. This round tests not only the mathematical capabilities of participants, but also their ability to work as a team to best their navigation capabilities. Through application of Trigonometric rules, participants will set off on a scavenger hunt. Participants will have to retrieve equations from the wild, making proper use of the distance and angle measurements as provided. Once all equations have been collected, participants will have to expend their mathematical abilities one last time, in order to reach the final destination (pun intended).

Note:

- The cumulative scores of teams in the first two rounds determine whether they proceed to the final round or not. The decision lies at the discretion of the category head.

Round 3: Blueprint Battles**ROUND DURATION: -****DELEGATE CAP: 3**

One may attribute many characteristics to Mathematics, but useless is not one of them - unless the Math is simply theory with no application. Applied Mathematics is arguably the most important field of study given its existence, essentially, everywhere. In the final round, teams will be provided real scenarios with mathematical applications, and they will be required to present their approach and reasoning in the provided solutions. The problems will be based on structural stability and integrity. Teams will be evaluated on the depth of content, mathematical accuracy, and effective elimination of threatening variables.

Note:

- Though only 3 participants are allowed to present their research in the final round, multiple may work on the solution.
- Mathematical formulae and/or online simulations of the solutions are **not** required. Participants may include said components, but not presenting them will not lead to reduction in points.

Additional rules:

- Each team begins with 120 points. In the first two rounds, teams are allowed to use the points for:
 - A hint: 20 points
 - Extra time: 40 points
 - Single use of AI: 50 points
 - Answer verification: 50 points
- Unused points will be added to the team's final score.
- The winning and runners-up teams will be decided via their cumulative scores.

- Overall, participants should have knowledge of:
 - Number Theory
 - Patterns
 - Functions
 - Trigonometry and Bearing calculations
 - Polar Coordinates
 - Linear Algebra

Judging Criteria:

TOTAL POINTS: 300

- Round 1, total points: 100
 - Decryption is worth 60% of the points.
 - Final answers are worth 40%, if all are correct.
- Round 2, total points: 100
 - Retrieval of an equation yields 10% points.
 - Accurate reading and mapping of bearings and distances, and elimination of incorrect candidate locations are worth 20% points total.
 - Accurate use of mathematical rules along with time taken to complete the hunt is worth between 10% to 50% points.
- Round 3, total points: 100
 - Depth of content is worth 25% points
 - Accuracy of solution is worth 25% points
 - Practicality of solution is worth 25% points
 - Clarity of Concepts is worth 25% points